Suggested problems: 8, 9, 13

Problem 1: Suppose that a point (X, Y) is chosen at random from the unit square with $0 \leq x \leq 1$ and $0 \leq y \leq 1$. Compute $E(X^2 + Y^2)$.

Problem 2: (Example 4.1.14) The current value of a single share of a stock is $S_0 = \$200$. In one year, the price of the stock will either be $S_1 = \$260$ or $S_1 = \$180$, the former with probability q. Suppose that you may invest money at a risk-free annual rate of 4%. You have decided to purchase an *option contract* that allows you the right (but not the obligation) to buy the stock at a price of $K = \$200$ in one year. This contract costs money, and the goal of this problem is to compute its price.

(a) First we compute q as a *risk-neutral probability*. This means that purchasing one share of the stock right now for $S_0 = \$200$ will be equivalent (in expectation) to investing \$200 at the 4% risk-free rate. Compute q. (With q chosen this way, the price S_0 of the stock is called its *risk-neutral price*.)

(b) Suppose you purchase the option contract. If, in one year's time, the price of the stock is \$260, then you purchase the stock at $K = \$200$, then turn around and sell it for \$260 and pocket \$60 in profit. If the price of the stock is \$180, you let the contract expire worthless. So, the payoff of the stock is the random variable $\Phi(S_1)$, where

$$\Phi(s_1) = \begin{cases} 60 & : s_1 = 260, \\ 0 & : s_1 = 180. \end{cases}$$

Compute the risk-neutral price of the option contract.